

Exercise

1. What are the **7 basic principles** of object orientation? Provide a brief description for each principle.

1. Divide-and-Conquer - break problems into small, manageable tasks
2. Encapsulation and Modularity - Each object knows how to solve its task and has info it needs
3. Public Interface - Each object should present a clear public interface that determines how other objects will be used
4. Information Hiding - shield users from unnecessary details of how it performs its role
5. Generality - Objects design to solve a kind of task rather than singular task
6. Extensibility - object functionality should designed that can be extended to carry out more specialized tasks
7. Abstraction - Simplify complex systems by showing only essential features and hiding unnecessary details

2. State the **benefits** of OOP.

- Save development time and cost by **reusing code**
 - Once an object class is created it can be used in other applications
- Easier **debugging**:
 - Classes can be tested independently
 - Reused objects have already been tested